GlyGen: computational and informatics resources for glycoscience

Search within citing articles

The glycoconjugate ontology (GlycoCoO) for standardizing the annotation of glycoconjugate data and its application

I Yamada, MP Campbell, N Edwards, LJ Castro… - …, 2021 - academic.oup.com

Recent years have seen great advances in the development of glycoproteomics protocols and methods resulting in a sustainable increase in the reporting proteins, their attached …

☆ Save Cite Cited by 18 Related articles All 15 versions

Semantic annotation of Glycomics and Glycoproteomics methods

W Wang, V Kuzyk, GSM Lageveen-Kammeijer… - …, 2026 - academic.oup.com

Glycomics and glycoproteomics represent the systematic exploration of glycan structures and glycoprotein compositions within biological systems, aiming to elucidate their roles in …

☆ Save Cite Related articles All 8 versions

Exploring protein glycosylation across GlyGen, UniProtKB, and related knowledgebases for data integration

J Vora, K Warner, P Vasudev, S Orchard, S Kulkarni… - Database, 2026 - academic.oup.com

Protein glycosylation plays a crucial role in cellular function, protein stability, and disease mechanisms, making the accurate annotation and integration of glycobiology data essential …

☆ Save Cite Related articles All 3 versions

Tools for generating and analyzing glycan microarray data

AY Mehta, J Heimburg-Molinaro… - Beilstein journal of …, 2020 - beilstein-journals.org

Glycans are one of the major biological polymers found in the mammalian body. They play a vital role in a number of physiologic and pathologic conditions. Glycan microarrays allow a …

☆ Save Cite Cited by 18 Related articles All 8 versions »

MGAT3 and MGAT5 overexpression alters the protein cargo of extracellular vesicles released by metastatic melanoma cells

M Wilczak, M Surman, U Jankowska… - Biochemical and …, 2025 - Elsevier

Extracellular vesicles (EVs) are potential non-invasive diagnostic, prognostic and therapeutic tools. Additionally, they are important contributors to tumorigenesis …

☆ Save Cite Cited by 2 Related articles All 6 versions

Editor's Choice GlycoEnzDB: a database of enzymes involved in human glycosylation

Y Zhou, V Ghosh, S Venkatesan, S Sriram… - …, 2025 - academic.oup.com

The glycan distribution on cells is governed by the activities of different families of enzymes that are together called “glycoEnzymes.” These include~ 400 gene products or 2% of the …

☆ Save Cite Cited by 2 Related articles All 12 versions

Dealing with the ambiguity of glycan substructure search

V Daponte, C Hayes, J Mariethoz, F Lisacek - Molecules, 2021 - mdpi.com

The level of ambiguity in describing glycan structure has significantly increased with the upsurge of large-scale glycomics and glycoproteomics experiments. Consequently, an …

☆ Save Cite Cited by 14 Related articles All 14 versions »

Bridging glycoinformatics and cheminformatics: integration efforts between GlyCosmos and PubChem

T Cheng, T Ono, M Shiota, I Yamada… - …, 2023 - academic.oup.com

Abstract The GlyCosmos Glycoscience Portal (https://glycosmos. org) and PubChem (https://pubchem. ncbi. nlm. nih. gov/) are major portals for glycoscience and chemistry …

☆ Save Cite Cited by 9 Related articles All 9 versions

Communicating regulatory high-throughput sequencing data using BioCompute Objects

CHS King IV, J Keeney, N Guimera, S Das, M Weber… - Drug discovery today, 2022 - Elsevier

This project demonstrates the use of the IEEE 2791–2020 Standard (BioCompute Objects [BCO]) to enable the complete and concise communication of results from next generation …

☆ Save Cite Cited by 13 Related articles All 7 versions

Integrating plasma protein-centric multi-omics to evaluate the causal effect of glycosylation on the risk of cancer

Z Geng, K Qi, L Yu, Y Ruan, J Gu, S Song - Glycoconjugate Journal, 2026 - Springer

Glycosylation is a crucial post-translational modification, and numerous studies have reported its significant role in cancer progression. Nevertheless, no study has …

☆ Save Cite Related articles All 3 versions